

Networks: flows

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Solution of the practice quiz

1. The cut is defined as $\Gamma = (\mathcal{M}, \mathcal{N} \setminus \mathcal{M})$ with $\mathcal{M} = \{v_0, v_1, v_2, v_3, v_6\}$ and $\mathcal{N} \setminus \mathcal{M} = \{v_4, v_5, v_7, v_8\}$.
 $\Gamma^{\rightarrow} = \{(v_3, v_4), (v_6, v_5), (v_6, v_8)\}$ is the set of forward arcs.
 $\Gamma^{\leftarrow} = \{(v_5, v_3)\}$ is the set of backward arcs.

2. Recall that the flow through the cut Γ is defined as

$$\mathcal{X}(\Gamma) = \sum_{(i,j) \in \Gamma^{\rightarrow}} x_{ij} - \sum_{(i,j) \in \Gamma^{\leftarrow}} x_{ij}.$$

Then we obtain

$$\mathcal{X}(\Gamma) = (-3 + 4 - 5) - (6) = -10.$$

The interpretation is that there is a total of 10 units of flow transported from the set $\mathcal{N} \setminus \mathcal{M}$ to the set $\mathcal{M}$.